



Research on Methodology of Intelligent Traffic Accident Detection based on Enhanced YOLOv8 Algorithm

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ABSTRACT

In response to the problem of complex visual feature categories and sparse datasets for related refined categories in traffic accidents, this article has created a traffic accident target detection dataset, the AccOD dataset, which is divided into five categories. This dataset can serve as a benchmark for training and evaluating traffic accident target detection algorithms. Aiming at the real-time, accuracy, and intelligence issues of traffic accident detection, an improved traffic accident target detection network based on YOLOv8 is proposed. This network replaces a more efficient backbone network structure and improves the downsampling module that integrates different scale features in the head network, reducing the number of model parameters and computational complexity while improving the detection accuracy of the model. The C2f module is improved to balance various contextual information, further enhancing the detection ability of the model. The experiment showed that the improved model achieved a 4.75% improvement in mAP0.5 and 4.58% improvement in mAP0.5:0.95 on the AccOD dataset compared to YOLOv8n, and a 2.14% improvement in mAP0.5 and 6.23% improvement in mAP0.5:0.95 on the open source ACD dataset. This algorithm improves the detection accuracy of traffic accidents and maintains the lightweight of the model, providing an effective method for intelligent traffic accident detection.

CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; Computer vision.

KEYWORDS

Traffic accident detection, YOLOv8, Object detection, Context information, Dataset, Lightweight

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1 INTRODUCTION

With the rapid development of China's economy, the number of cars held has been increasing year by year, and traffic accidents have also shown a high incidence trend. According to data from the China Bureau of Statistics, the number of traffic accidents in the past five years has been over 240 thousand, with an annual death toll of over 60 thousand [1], causing serious losses to the country and people. Timely detection of accidents can save lives, reduce losses, and avoid secondary accidents to the greatest extent possible, which is crucial for improving traffic management, public safety, and rescue response. The traditional manual detection method is time-consuming and labour-intensive. The intelligent traffic accident detection method based on computer vision has the advantages of high efficiency, accuracy, and strong automation adaptability, which is of great significance for the timely detection and rescue of accidents and promotes the development of intelligent traffic management systems.

In the field of object detection, deep learning models have achieved significant results, but these methods often encounter difficulties in processing traffic accident images. There are two main types of difficulties, one is in terms of data. Due to the danger and rarity of traffic accidents, obtaining large-scale and detailed traffic accident datasets is very difficult. This results in a lack of large-scale and diverse datasets in the field of traffic accident target detection. Moreover, there are various types of targets involved in traffic accidents, with different shapes, colours, and sizes. It is difficult to classify the types of accidents based on visual features because even accidents of the same type in semantics will

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have different feature representations due to changes in the accident subject, vehicle status, and other factors. A scientific accident classification method is particularly important for the accurate detection of targets involved in accidents. The research on detection methods and algorithms for these specific types of accidents is of great significance for preventing traffic accidents and reducing their consequences.

Another challenge is in feature extraction, where the visual features of traffic accidents include complex factors such as various objects, vehicle collisions, and environmental backgrounds, resulting in a diversity of categories and height changes in images. The accident scene is variable, including various weather conditions, lighting, and road conditions. The mutual occlusion and different motion states of targets, as well as varying degrees of damage, can also affect the appearance and shape of targets, which increases the difficulty of target detection.

In response to the above difficulties, the main contributions made in this article are as follows:

- (1) This article summarizes various features in traffic accident scenarios, divides accident targets into five relatively independent categories, and annotates them on 7778 images to construct a traffic accident target detection dataset - the AccOD dataset.
- (2) This article replaces the backbone network of YOLOv8 with the EfficientViT network, which can better understand the relationship between different subjects in traffic accidents, focus on key features of accidents, and balance detection accuracy and speed.
- (3) This article introduces an improved lightweight downsampling module and a C2f_MLCG module that integrates multiple contextual information in the head network, which further reduces the model's parameter and computational complexity while maintaining detection accuracy.

2 RELATED WORK

The traditional visual accident detection technology based on image processing and computer vision technology mainly includes background modelling and feature detection methods. The methods based on background modelling are divided into the frame difference method [2] and the background subtraction method [3]. Based on inter-frame difference, the target is detected by calculating the difference between adjacent frames, and based on background subtraction, the target is detected by comparing the difference between each frame and the background image.

The second is based on feature detection, including traditional object detection and object tracking methods. Traditional object detection uses features such as SIFT [4], HOG [5], and Harr [6] to extract features, and uses SVM [7] and AdaBoost [8] for recognition. Traditional target tracking includes template matching based mean shift algorithm [9] and CAMShift algorithm [10], as well as the Lucas Kanade algorithm [11] and SURF algorithm [12] based on feature point tracking. However, traditional methods have certain shortcomings. Based on the sensitivity of inter-frame difference to lighting changes, background subtraction of long videos may cause background drift. Traditional object detection has high subjectivity and incomplete feature extraction. Processing big data requires

computing resources, and machine learning requires a large amount of annotated data. Traditional target tracking has the problem of error accumulation, making it difficult to truly achieve vehicle collision detection.

The application of object detection technology in traffic accident detection has made significant progress. Early studies mainly improved detection accuracy by improving YOLO-based models, but data size and model training were limited. For example, Gour et al. [13] implemented accident detection systems through optimized YOLO algorithms. Pillai et al [14] proposed a three-stage vehicle accident automatic detection system framework with low computational cost and ideal performance, using the Mini YOLO model for detection. Later, research began to integrate object detection and other technologies such as target tracking to establish an end-to-end recognition framework, such as the YOLO-CA model proposed by Tian et al [15], which improved real-time performance and accuracy through multi-scale feature fusion and dynamic weight adjustment. Xia et al [16] improved the YOLOv5l model to enhance its accuracy and efficiency. Other studies have combined object detection with other algorithms, such as Chakraborty et al [17] proposing a semi-supervised parameter estimation method that utilizes the YOLOv3 object detection algorithm and SORT algorithm for vehicle trajectory extraction and tracking. Lee et al [18] constructed a real-time traffic accident notification application using YOLOv3. Ghahremannezhad et al [19] proposed a real-time accident detection framework for traffic intersections. Basheer et al [20] constructed a real-time traffic accident detection and alarm system using YOLOv5 and DeepSORT trackers. Ijjina et al [21] demonstrated through experiments that a traffic accident detection framework based on Mask R-CNN and motion feature analysis can also play a good role in complex scenes. However, overall, the methods of target tracking and trajectory analysis have certain limitations in accuracy and real-time performance due to occlusion and historical trajectory requirements.

Object detection technology has made progress in traffic accident detection, but challenges remain. These include adapting to diverse and complex traffic scenes, accurately analyzing target motion and behaviour, and the demanding task of dataset annotation. Integrating object detection with other technologies is crucial for efficient and automatic traffic accident detection.

3 THE PROPOSED METHOD

3.1 Model Architecture

The overall structure of the model is shown in Figure 1: Overall structure of the model. In this chapter, the backbone network composed of the original multi-layer downsampling and C2f module on the left has been replaced by the EfficientViT [22] feature extraction network. This network adopts a lightweight network structure and utilizes an efficient Transformer to effectively fuse multi-scale features and improve feature extraction capabilities, achieving a good balance between speed and accuracy. The detection head and pyramid feature fusion section on the right side of the image are collectively referred to as the head network. This model introduces a lightweight and informative downsampling module in the downsampling process of the head network, integrating features of

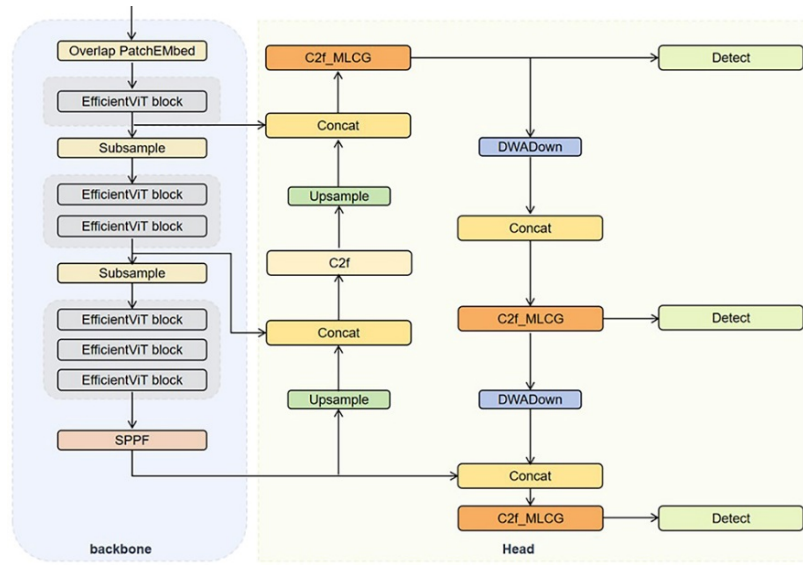


Figure 1: Overall structure of the model.

different semantic and spatial scales to reduce information loss during the downsampling process. In addition, this model replaces the C2f module in the head network with the C2f_MLCG module that integrates various contextual information, further improving the detection accuracy of the model and reducing the number of model parameters, making it more suitable for the real-time requirements of traffic accident detection.

3.2 Efficient Backbone Network

With the in-depth research of ViT models, a new series of models called EfficientViT has emerged to address the computational cost of ViT in real-time applications. EfficientViT is a lightweight visual Transformer (ViT) model aimed at achieving efficient real-time inference. The n model of YOLOv8 is the smallest model in the YOLOv8 series, and in order to maintain its lightweight, EfficientViT sacrifices a certain feature extraction ability. EfficientViT adopts an innovative module design, including a sandwich layout and cascaded group attention module. This model improves detection accuracy while maintaining lightweight, and its structure is shown in Figure 2.

Each EfficientViT block adopts a sandwich structure, consisting of N feedforward neural networks (FFNs), one cascaded group attention (CGA), and then N FFNs. This design helps to reduce the memory access time of multi-head self-attention (MHSA) computation, thereby improving inference speed.

Cascade Group Attention (CGA) is a key component of EfficientViT, which is an improvement on the standard multi-head self-attention mechanism (MHSA). The main idea is to group the computation of attention and share weights within each group. In addition, EfficientViT conducted parameter importance analysis to rearrange the composition of the network, giving smaller dimensions to Q and K, adjusting the dimension of V, and using FFN with an expansion ratio of 2 to achieve more efficient resource utilization.

This study requires real-time detection of traffic accidents, and the model has superior performance and speed in real-time inference tasks. Therefore, we replaced the backbone network of YOLOv8 with the smallest EfficientViT_M0, which has different head widths, depths, and quantities at each stage. In the forward propagation algorithm, the output of each stage is stored in a list of outputs as the output of the model, and the outputs of the three stages are input as feature maps of different scales into the head network of YOLOv8.

3.3 Lightweight Downsampling Module

The downsampling convolution module in the YOLOv8 head network consists of a convolutional layer, batch normalization layer, and SiLU activation function. This module is used to reduce the resolution of input feature maps while increasing the number of channels for output feature maps. However this method of reducing the size of feature maps by sampling with a fixed window size inevitably suffers from information loss and uneven sampling. Therefore, this section proposes the DWADown module to optimize the downsampling process of the neck network. The structure of this module is shown in Figure 3.

The DWADown module performs an average pooling operation on the input feature map and then divides it into two parts using the Chunk operation. For each part, different convolution kernels are used for processing, which is either directly using depthwise separable convolutions or a combination of max pooling and depthwise separable convolutions. This design reduces the number of parameters required for convolution operations and improves computational efficiency.

During the downsampling process, the DWADown module integrates average pooling and maximum pooling operations. Average pooling helps to extract overall features and reduce complexity, while maximum pooling highlights salient features and improves positional invariance. By combining these pooling operations, the

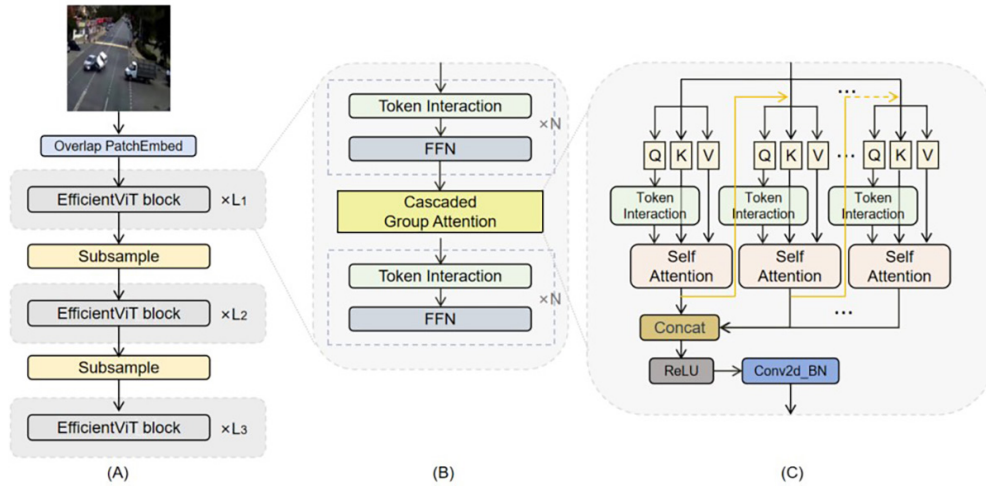


Figure 2: EfficientViT network structure (A) network structure; (B) The sandwich structure of each EfficientViT block; (C) Cascade attention structure.

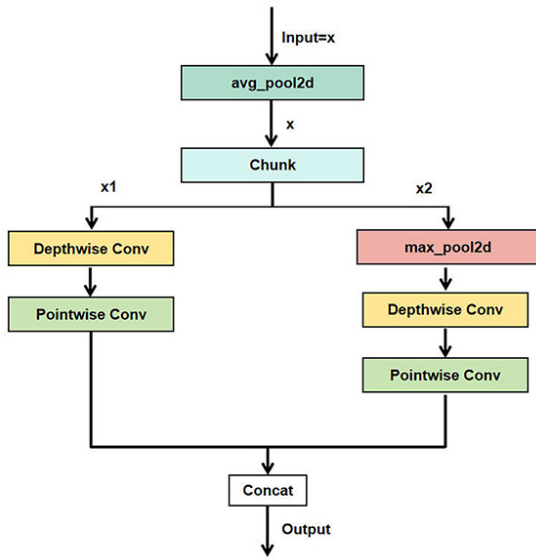


Figure 3: DWADown Module.

DWADown module preserves the detailed information in the feature map and integrates features of different semantic and spatial scales. This enhances the model’s ability to perceive targets while maintaining a lightweight design, improving detection accuracy.

3.4 Multi-Layer Context-Guided Module

YOLOv8 utilizes the C2f module to fuse low-level feature maps containing more detailed information with high-level feature maps containing more semantic information. The key to obtaining high-level feature maps is the bottleneck part. In order to better obtain the semantic information of the context, this module refines the

extraction operation of local information on the basis of the context-guided block, forming a multi-layer context-guided block (MLCG) that integrates local, surrounding, and global context information. The method of information extraction and fusion is shown in Figure 4.

The main idea of this module is to enhance the expression ability of features by modelling the spatial coordinates of feature maps and introducing spatial coordinate information. In traditional convolutional neural networks, each position of the feature map is treated equally regardless of which part of the image it is in. However, when it comes to object detection tasks, visual information from different positions may have different importance. The goal of coordinate attention is to capture the importance of different positions in the feature map and apply these weights to the corresponding features.

The feature maps, weighted by coordinate attention, are processed in parallel operations to extract local and surrounding context information. Local context is obtained through standard 3x3 convolutions while surrounding context is captured using dilated convolutions with a dilation rate of 2. The resulting features are concatenated and further enhanced through batch normalization, parameterized ReLU operations, and global context refinement. This enables the integration of local and surrounding contextual information, enhancing the overall representation.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup, Parameters, and Evaluation Metrics

The experiments in this study were conducted on a computer running Ubuntu 20.04.4 operating system. The computer was equipped with an RTX 2080 Ti graphics card (11GB VRAM) and an Intel(R) Xeon(R) Platinum 8255C processor with 12 cores operating at a base frequency of 2.50 GHz. The system had a memory capacity of 50GB. The deep learning framework used for the experiments was

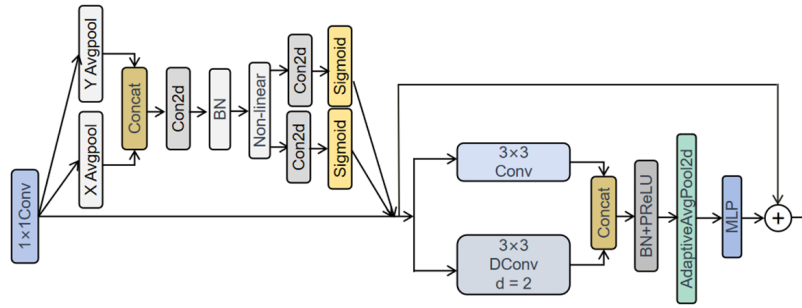


Figure 4: Multi-Layer Context-Guided Module.

PyTorch version 1.11, running on a CUDA 11.3 environment, with Python version 3.8.10.

Specific experimental parameters were set as follows: The input image size was set to 640×640. The training process consisted of 300 iterations, with a batch size of 16 images per iteration. Stochastic Gradient Descent (SGD) optimization was used, with an initial learning rate of 0.01 and a momentum of 0.9. Weight regularization was applied using an L2 penalty with a coefficient of 0.0005. The training process was stopped if the accuracy did not improve for 50 iterations. Data loading was accelerated using 8 threads. The model's weights were initialized with a pre-trained model, YOLOv8n.pt, and the WIoU loss function was used.

The evaluation metrics used in this study were precision, recall, and average precision. Mean Average Precision (mAP) at IoU thresholds of 50% (mAP50) and 50% to 95% (mAP50-95) were used to measure the average precision across different categories. Precision measures the accuracy of the model by calculating the ratio of true positive predictions to all positive predictions. Recall measures the ability of the model to correctly detect positive samples by calculating the ratio of true positive predictions to all actual positive samples.

4.2 DataSets

4.2.1 Accident object detection dataset. Due to the need for real-time detection of traffic accidents in video streams, most of the self-built datasets in this study were extracted from traffic accident videos and collected traffic accident images from different regions, environments, and types through various sources. Road scenes include urban roads, highways, rural roads, and other scenes, each scene also includes accidents under different lighting, weather, and other environmental conditions. In addition to searching various online resources, the collection of the dataset also utilized several publicly available traffic accident datasets, such as the CADP dataset and TAD dataset, to increase the diversity and coverage of the dataset. This dataset has undergone strict screening among the collected images, excluding images unrelated to traffic accidents, such as landscape photos, road signs, etc. In addition, we also conducted quality checks on the images and removed blurry, over-exposed, or other quality issues to ensure that the image quality in the dataset meets the requirements.

The annotation tool used in the annotation process is LabelImg, which annotates the traffic accident targets in each image and

assigns a category label to each target. This dataset defines five categories: multi-vehicle collision, accident four-wheel vehicle, fire, accident two-wheel vehicle, and accident pedestrian. The calibration standards for these five categories are as follows: for multi-vehicle collision categories, label two or more vehicles that collide as one target. If they separate after the collision, we will label them separately as accident four-wheeled vehicles; The four-wheeled vehicles involved in the accident, including large trucks, buses, and ordinary cars, exhibit abnormal conditions such as overturning, deformation, or collision with obstacles; The types of ignition include open flames and thick smoke. If there are multiple sources of ignition, label multiple ignition targets separately; Abnormal conditions such as overturning, deformation, and collision of two-wheeled vehicles in accidents, including motorcycles, electric vehicles, and bicycles; Most accident pedestrians are situations where pedestrians or drivers of two-wheeled vehicles are knocked down or thrown away. Figure 5 shows the labelling of each category. This dataset is named Accident Object Detection Dataset (AccOD), which contains 7778 images and a total of 13637 labels. The ratio of the training and validation sets is 7:3.

4.2.2 Accident object detection dataset. This dataset is sourced from the open-source dataset website [23], which consists of 16114 images and 21452 tags. There is only one type of tag, which is the accident tag. The types of accidents include collision, overturning, fire, damage and other abnormal states, and the accident subjects also include large trucks, buses, ordinary cars, two-wheeled vehicles, etc.

4.3 Comparison Experiment

In order to verify the performance of the improved model in traffic accident target detection, a comparative experiment was conducted on the AccOD dataset and CAD dataset under the same configuration, and compared with many lighter models. These two datasets respectively verified the model's ability to detect multiple and single targets in traffic accident scenes. Most of the models in the comparative experiments are YOLO series, which transform the object detection task into a single regression problem and adopt an end-to-end training method. They perform well in detection speed and accuracy, especially when processing high frame-rate videos or real-time stream data, they can maintain high frame rates and meet real-time requirements.

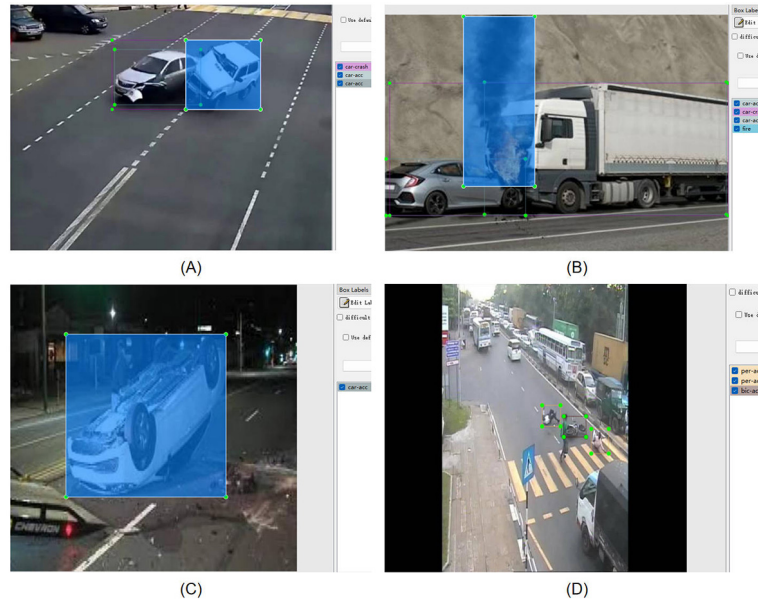


Figure 5: Using the Labeling to annotate schematic diagrams of various categories in the AccOD dataset.

Table 1: Comparative experimental results on the AccOD dataset.

Models	FLOPs(G)	Params(M)	Precision(%)	Recall(%)	mAP@0.5:0.95(%)	mAP@0.5(%)
YOLOv5n	4.2	1.8	83.01	75.83	56.80	82.30
YOLOv5s	16.5	7.22	84.57	77.03	59.81	83.23
YOLOv5m	48.3	20.88	87.25	77.96	61.58	84.36
YOLOv6n [83]	11.9	4.2	79.82	70.56	55.82	77.10
YOLOv6s	44.2	16.31	83.11	74.19	59.65	81.55
YOLOv7-tiny [84]	13.1	6.01	80.78	72.27	55.18	78.25
YOLOv7	103.5	36.49	84.10	80.18	56.43	84.27
RT-DETR-R18 [85]	47.9	20.05	81.82	73.66	56.74	80.56
RT-DETR-R50	136.1	42.11	85.37	76.24	59.03	84.52
RT-DETR-L	110.0	32.73	85.25	75.48	58.86	84.34
YOLOv8n [86]	8.2	3.10	81.19	73.81	58.07	79.98
YOLOv8s	28.7	11.1	86.09	74.45	60.75	83.10
Ours	8.6	3.4	87.73	79.25	61.65	84.73

The comparative experimental results on the AccOD dataset are shown in Table 1. In this chapter, the model achieved optimal accuracy while maintaining a lightweight parameter size of 3.4M and GFLOPs of 8.6. Compared to the YOLOv8n model with the same volume, it increased mAP by 4.75 percentage points at an IoU threshold of 50%. The backbone network of YOLOv5 mainly consists of Focus, BottleneckCSP, and SSP networks, which mainly include modules such as Focus, Conv convolution blocks, BottleneckCSP, and SSP. Among the three relatively lightweight models in the YOLOv5 series, the n model is the lightest among all comparative experiments, with only a parameter size of 1.8M and 4.2 GFLOPs. While lightweight, its accuracy is slightly lower than that of the s model and m model, which is 2.43 percentage points lower than the model proposed in this chapter.

The YOLOv6 [24] series algorithm has undergone many innovations in structure. It adopts a hybrid channel strategy to redesign a more efficient decoupling head, which reduces delay while maintaining accuracy and effectively alleviates the additional delay overhead caused by convolution in the decoupling head. In this section, we also selected two smaller YOLOv6 models, n and s. As the model size increases, the accuracy improves, but it has not yet reached the best accuracy. The s model is 3.18% lower than the model in this chapter, indicating that the v6 model performs worse than other models on this dataset. Most of the overall network architecture of YOLOv7 [25] comes from YOLOv5, which has been optimized and integrated with various previous techniques. However, it has also increased the size and computational complexity of the model. The FLOPs of the v7 model have exceeded 100G, and the accuracy

Table 2: Comparative experimental results on the ACD dataset.

Models	FLOPs(G)	Params(M)	Precision(%)	Recall(%)	mAP@0.5:0.95(%)	mAP@0.5(%)
YOLOv5m	48.3	20.88	93.75	92.97	82.40	95.79
YOLOv6s	44.2	16.31	92.65	92.53	83.32	95.08
YOLOv7	103.5	36.49	94.03	92.44	83.48	96.10
RT-DETR-L	110.0	32.73	93.95	92.79	82.94	96.23
YOLOv8n	8.2	3.10	91.89	91.28	77.82	94.43
YOLOv8s	28.7	11.1	93.36	92.03	81.86	95.53
Ours	8.6	3.4	94.49	92.89	84.05	96.57

Table 3: Ablation experiments of different modules on two datasets.

Branches	FLOPs(G)	Params(M)	mAP(AccOD) @0.5:0.95(%)	mAP(AccOD) @0.5(%)	mAP(ACD) @0.5:0.95(%)	mAP(ACD) @0.5(%)
v8n	8.2	3.1	57.07	79.98	77.82	94.43
+EfficientViT	9.5	4.0	60.57	83.05	81.96	95.57
+DWADown	8.0	2.8	57.63	80.28	78.63	94.68
+C2f_MLCG	7.6	2.7	57.93	80.58	78.84	94.98
+DWADown+C2f_MLCG	7.3	2.5	58.12	81.03	79.61	95.21
+EfficientViT+DWADown	9.2	3.8	60.46	83.14	82.46	95.87
+EfficientViT+C2f_MLCG	8.8	3.6	61.04	84.12	83.57	96.04
Ours	8.6	3.4	61.65	84.73	84.05	96.57

of the lightest tiny model is poor. In the case of larger parameters and computational complexity than the model in this chapter, the mAP with an IoU threshold of 50% is 7.63% lower.

The RT-DETR [26] series model has designed an efficient hybrid encoder that efficiently processes multi-scale features through decoupling intra-scale interactions and cross-scale fusion. It also utilizes IoU perception to optimize decoder query initialization. The R18 and R50 of RT-DETR are different backbone networks, namely ResNet18 and ResNet50, while the L version's backbone network is HGNetv2. Although the detection accuracy of the R50 and L versions is close to that of the models in this chapter, the parameter and computational complexity of this series of models are relatively large. Finally, the baseline model YOLOv8 series [27] in this chapter adopts advanced and lightweight backbone and head networks. The n model with the smallest scaling ratio has only 3.1M parameters, but the detection accuracy is not high. The improved model in this chapter has an accuracy of 1.63% higher than the larger s model.

This section also conducted comparative experiments with relevant algorithms on open-source vehicle accident datasets. Among the same series of algorithms except YOLOv8, the algorithms with higher accuracy were selected from the AccOD dataset. The comparison results are shown in

Table 2. Because this dataset only has one accident category, the percentage points exceeding the relevant algorithms in accuracy are not as high as on the traffic accident target detection dataset, but it also achieves the best results. Moreover, it has much smaller parameters and computational complexity than algorithms with similar accuracy. Compared with baseline algorithms, it has improved accuracy, recall, IoU threshold from 50% to 95%, and threshold of 50% by 2.6%, 1.61%, 6.23%, and 2.14%, respectively.

4.4 Ablation Experiment

In this study, the baseline model used for evaluation was the YOLOv8n model. Evaluate the performance of each module in the fusion model under the same experimental environment and parameter settings. The evaluation indicators include the number of parameters, computational complexity, and mean accuracy (mAP) accuracy.

From Table 3, it can be seen that replacing the backbone network will significantly improve accuracy. The module improved two accuracy indicators by 3.5% and 3.07% respectively on the five types of accident object detection dataset, and improved by 4.14% and 1.14% respectively on the two detection accuracy indicators for single accident categories. Considering the characteristics of the traffic accident dataset used in this study, it typically includes various complex scenes and objects that differ in size, shape, posture, and occlusion. For example, the difference between vehicle collisions and congestion caused by oncoming vehicles is minimal, and the deformation of the accident vehicle is also different. In addition, there may be significant size differences between vehicle collisions and accidents involving pedestrians. Therefore, EfficientViT utilizing the Transformer architecture can use multi-head self-attention mechanisms and cascaded group attention modules to capture relationships and contextual information between objects. This helps to improve detection performance in complex scenes with different targets.

In addition, compared to single-class datasets, the greater improvement of multi-class datasets at a 50% IoU threshold indicates that this module can better utilize inter-class relationships and contextual information, thereby improving overall object detection performance. The greater improvement at higher IoU thresholds is

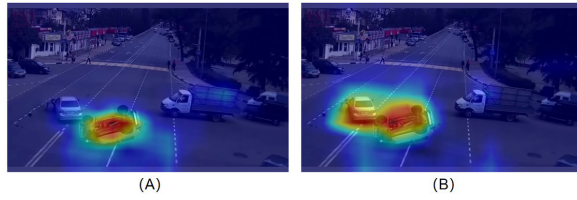


Figure 6: Heatmap comparison. (A) Baseline model visualization. (B) Improved model visualization.

due to stricter overlap requirements between detected boxes and ground truth boxes, thereby reducing false positive rates.

The table also shows the increase in the number of parameters and computational complexity of the new backbone network. However, the downsampling module in the head network and the C2f module that combines various contextual information effectively reduce the number of parameters and computational complexity. From the table, it can be seen that the DWADown module reduced the parameter count by 0.3M and computational complexity by 0.2G, while the C2f-MLCG module reduced the parameter count by 0.3M and computational complexity by 0.6G. When these two modules work together, the model becomes the lightest, and the semantic and contextual information in the high-level feature map complement each other. This is particularly beneficial in complex traffic accident scenarios, where local contextual information helps the network better understand the subjects involved in the accident and distinguish between different objects and surfaces, such as different types of damaged or colliding vehicles and falling pedestrians. Surrounding contextual information provides insight into the relative position of accident targets and surrounding information, while global contextual information refers to the environmental information of the entire traffic scene, such as vehicle density and layout after an accident. Therefore, it is necessary to combine contextual information in the feature extraction process. These two modules also contribute to improving detection accuracy.

The experimental results indicate that when these two lightweight modules are combined with the EfficientViT backbone network, they achieve better performance compared to using the EfficientViT network alone. The experiment shows that when the three modules work together, the model achieves the highest accuracy.

The qualitative analysis of the model's improvement is illustrated in Figure 6. In the figure, (A) depicts the visualization heatmap of the baseline model, while (B) presents the visualization heatmap of the improved model. It is evident that although the baseline model can identify the overturned vehicle, it overlooks the damage to the vehicle on the left side and fails to capture the collision relationship between the two vehicles. In contrast, the improved model successfully detects the minor damage to the vehicle on the left side and accurately captures the collision relationship between the two vehicles.

5 CONCLUSION

This paper presents an efficient traffic accident object detection network that replaces the backbone network of YOLOv8 with an efficient vision transformer network. This replacement allows the

network to capture relationships among different entities in traffic accident scenarios and focus on key regions of accidents, resulting in improved detection accuracy with minimal computational and operational complexity. Furthermore, the paper introduces the lightweight downsampling module, DWADown, and the C2f_MLCG module, which integrates local, contextual, and global information, reducing the model's size while improving detection accuracy. To address the lack of fine-grained traffic accident datasets, the paper constructs the AccOD dataset, divided into five accident categories. Comparative and ablation experiments on this dataset and the CAD dataset demonstrate the superiority of the proposed algorithm, providing an effective approach for intelligent traffic accident detection.

Future work can involve collecting more fine-grained data for different traffic accident categories, enhancing the model's ability to extract accident information. Additionally, integrating other sensor data (such as radar and LiDAR) can enable multimodal data fusion, leading to a more comprehensive understanding of traffic accident scenes.

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